Total Printed Pages: 02 Roll No. :.....

ECA-101
Examination – Dec- 2023
B.Tech. I Sem : Electronics and Communication
Basic Electronics

Time : 2 Hrs Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	A semiconductor diode is used as? (a) Amplifier (b) Rectifier	02	1,2	1
	(b)	For faster switching which type of diode is used? Justify your answer.	02	3,6	4
	(c)	Compare Zener and avalanche breakdown.	03	3,4	2
		OR			
		Draw the circuit diagram of half wave rectifier.	03	3,4	2
	(d)	With the help of VI Characteristics explain the working of Zener diode.	05	3,4	3
		OR			
		Calculate the value of peak reverse voltage (P.I.V.) if the full-wave rectifier has an alternating voltage of 300 V.	05	3,6	4
Q.2	(a)	Identify the condition for a transistor to act as an amplifier. (i) The emitter-base junction is forward biased and the base-collector junction is reverse biased (ii) No bias voltage is required (iii) Both junctions are forward biased (iv) Both junctions are reverse biased.	02	1,2	1
	(b)	In the active state, the emitter-base junction has a higher resistance than the collector-base junction. (i) True (ii) False	02	3,4	2
	(c)	Explain base width modulation.	03	3,6	3
		OR			
		Derive the relation between alpha and beta.	03	3,4	2
	(d)	Sketch the Circuit diagram of a NPN common base configuration explain its working and VI characteristics	05	3,4	2
		OR			
		With the help of circuit diagram explain voltage divider transistor biasing.	05	3,4	3

Q.3	(a)	If the operating point of transistor changes the h-parameter also (i) Also Change (ii) Do not change	02	1,2	1
	(b)	The dimension of hie parameter are (i) mho (ii) ohm (iii) Farad (iv) none	02	1,2	1
	(c)	Draw the hybrid equivalent circuit for common base transistor.	03	3,4	2
		OR			
		Draw the hybrid equivalent circuit for common emitter transistor.	03	3,4	2
	(d)	Derive the equations for current gain and voltage gain of a CE amplifier using h-parameters.	05	3,4	2
		OR			
		Give low frequency h-parameter model of BJT and explain the same.	05	3,6	3
Q.4	(a)	Name different types of coupling.	02	1,2	1
	(b)	Define multistage Amplifier.	02	1,2	1
	(c)	Derive the equation for the overall voltage gain of multistage amplifier in terms of the individual voltage gain.	03	3,4	2
		OR			
		Differentiate between cascade and cascode amplifier.	03	3,4	2
	(d)	Draw the circuit diagram of Darlington Amplifier and explain.	05	3,4	2
		OR			
		With the help of neat sketches explain RC coupled amplifier.	05	3,4	2
Q.5	(a)	How many stages of a tuned amplifier does a staggered tuned amplifier comprises of? (i) Single (ii) Double (iii) Dual (iv) Multiple	02	1,2	1
	(b)	Which of the following are the advantages of tuned amplifier? (i) Good SNR (iii) Greater selectivity (ii) Low power loss (iv) All the above	02	1,2	1
	(c)	Discuss crossover Distortion.	03	3,4	2
		OR			
		Discuss harmonic Distortion.	03	3,4	2
	(d)	Discuss the characteristics and response of double tuned amplifiers.	05	3,4	2
		OR			
		Differentiate A, B, AB and C type of power amplifier.	05	3,6	3

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

*****